Smart Collectorate

Taking a District Magistrate's office from paper files to an AI-enabled workflow, and getting people to use it

Pulse Rajan · Applied AI Engineer assessment · Creditfix Group · November 2026

The problem: district backlogs are growing

1 crore+

grievances on CPGRAMS in the last five years

157 → 15

days to dispose, central ministries, 2014 → 2025

82.1%

closed within the 21-day target in 2025, so about 1 in 5 is late

2.16 lakh

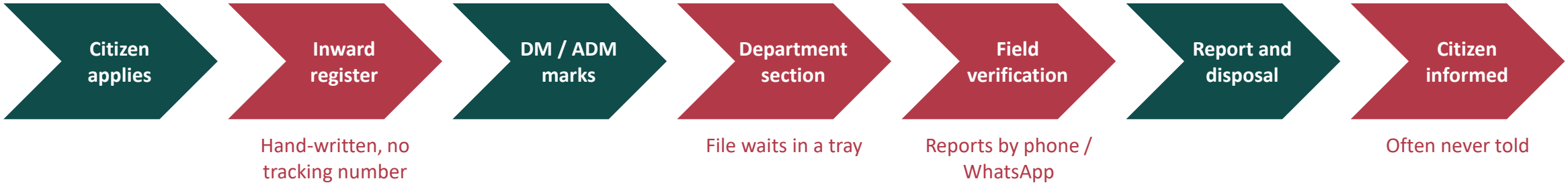
pending with States/UTs (June 2026), up ~26% since Dec 2025

In a typical district office (our working assumptions, measured in discovery)

- ~4,000 grievances and applications a month, ~30 days to dispose, 1 in 3 citizens come back only to ask for status
- Paper files and registers, follow-ups on phone and WhatsApp, no single owner per step, no live view for the DM

Sources: DARPG CPGRAMS monthly reports (PIB, 2026); Lok Sabha replies as reported by ThePrint and All India Radio News (2025–26); data.gov.in.

A. Discovery: map the workflow with its people



Who we talk to (25–30 interviews)

- DM, ADM, SDMs: which three numbers do you want every Monday?
- Clerks and inward desk: walk me through one file
- Field staff: how do reports reach you and go back?
- Citizens at the gate: how many visits did this take?

Assumptions we check, and how

- Mostly paper, mostly Hindi → count one month of the register
- ~10 categories cover 70% → hand-tag 300 past cases
- Data stays in India → confirm with NIC
- CPGRAMS / IGRS stays the system of record → check API access

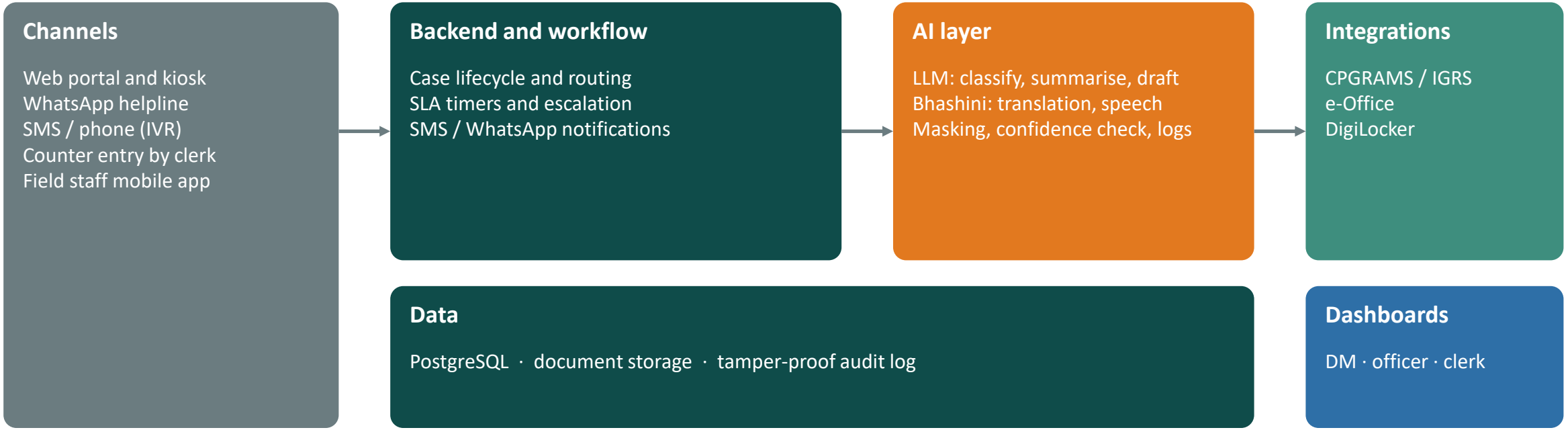
2–3 weeks · trace 50 real files with dates at every hand-off · rank steps by delay × volume

B. Seven use cases, and where AI is really needed

#	Use case	Why it matters	AI or rules?	Phase
1	Grievance intake and smart routing	Right department on day one; reads Hindi/English free text	AI	Pilot
2	QR code tracking on every file	No lost files; anyone can see who holds a file now	Rules	Pilot
3	SLA reminders and escalation	Files stop waiting in trays unnoticed	Rules	Pilot
4	Live dashboard for the DM	Backlog and overdue cases by department, today	Rules	Pilot
5	WhatsApp helpline (Hindi / English)	File and check status without visiting	AI	Rollout
6	Weekly review-meeting reports	Minutes instead of two days of compiling	Rules + AI	Rollout
7	AI assistant for officers	Summarise files, draft replies, find past orders	AI	Scale

Our rule: AI only where the input is messy language. Tracking, deadlines, alerts and dashboards are plain rules: cheaper, predictable, auditable. AI suggests; a person decides.

Architecture, and one grievance start to finish





1. Citizen files

Any channel, Hindi or English



2. Gets a number

Tracking ID by SMS



3. AI suggests

Unsure? A clerk confirms



4. Officer acts

Sees summary and deadline



5. Reminders

At 50% and 90%, then ADM



6. Closed

Citizen told and rates it

C. Roadmap: zero to district-wide in 12 months

Discovery

Weeks 1–3

- Workflow map, baseline
- Use cases signed off

Team: Lead, AI engineer, UX researcher

~8 person-weeks

Pilot

Weeks 4–14

- 2 departments, 1 tehsil
- Use cases 1–4 live
- 4 weeks parallel with paper

Team: Lead, 2 devs, AI engineer, QA, trainer

~45 person-weeks

Rollout

Months 4–6

- Whole district
- WhatsApp helpline
- Weekly reports

Team: Pilot team + 1 dev, 2 trainers

~70 person-weeks

Scale

Months 7–12

- Officer AI assistant
- Field app, integrations

Team: 2 devs, AI engineer, QA, DevOps

~80 person-weeks

Support

Ongoing

- Helpdesk and fixes
- Monthly AI review

Team: Support engineer, helpdesk

~1.5 people

Go / no-go at week 14: we scale only if the pilot targets are met. Office side: one nodal officer (~20% time) and a champion clerk per department.

What a successful pilot looks like

−40%

average time to dispose of a case

−30%

backlog older than 30 days

≥ 85%

AI routing accuracy (confirmed by officers)

≥ 80%

of officers using it every week

≥ 4 / 5

citizen rating of the service

0

lost files in pilot departments

All measured against the discovery baseline, for the same departments, after 10 weeks live.

D. Risks, and how we control them

Citizen data

- DPDP Act 2023: collect only what is needed
- Hosted in India (NIC / approved cloud)
- Encrypted, role-based OTP login, full audit log
- Aadhaar and phone masked before any AI call

AI mistakes

- AI suggests, an officer decides
- Under 70% confidence → a person checks
- Fixed department list; AI cannot invent one
- Accuracy and bias reviewed monthly

System and vendor

- AI down → routing falls back to the clerk
- Daily backups; works on slow connections
- CERT-In empanelled security audit
- Open source; data exportable any time

Staff adoption

- Hindi first, fewer steps than the register
- A champion clerk in each department
- 30-min desk-side training; paper kept 4 weeks
- DM reviews from the dashboard in meetings

E. The pitch, the objections, and the price

> In ten weeks, in two departments, every application gets a tracking number and reaches the right desk on day one, and you see every delayed file on one screen. Your staff still decide. Data stays in India. If delays don't fall by a third, you don't pay for the next phase.

"We already have a system." We fill gaps around CPGRAMS and e-Office, and feed data back.

"It is not secure." India hosting, OTP login, audit log, security audit.

"Staff won't use it." Hindi, fewer steps, champions; usage tracked weekly.

Pilot · 10–12 weeks

₹5–8 lakh

Fixed fee, 2 departments; next phase only if targets are met

Rollout · Module by module

₹20–30 lakh

Whole district, bought through GeM

Support · Yearly

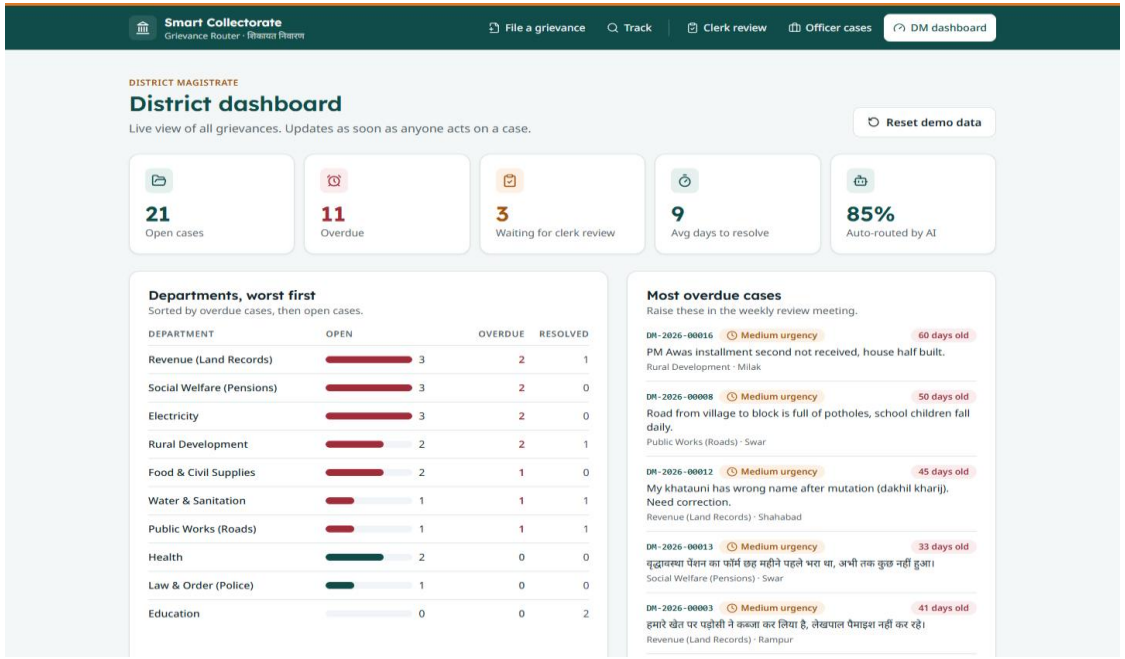
15–20% / yr

Of build cost: hosting, helpdesk, AI monitoring, training

Indicative figures, firmed up after discovery. Discovery can be free, to earn trust.

F. Working proof: AI grievance router, live

- Complaint in Hindi, English or a mix; phone and Aadhaar masked before the AI sees it
- AI suggests department, urgency and a summary, with a confidence score
- Under 70% confidence → clerk review queue; every decision logged
- Tracking ID, deadline clock, DM dashboard of overdue cases
- 20 test cases: Claude Sonnet 100% correct, keyword backup 90%, 0 wrongly routed



Live: smart-collectorate-grievance-router.vercel.app

Code: github.com/easycodewithme/Assignemnt-Automation

Next steps: 1. Approve a 3-week discovery (no cost) 2. Name a nodal officer and two pilot departments 3. Agree targets and the go / no-go date